Grade 1 Mathematics

Մաթեմատիկա — 1-ին դասարան

COURSE CODE	MTH-G1
PROGRAM	Elementary Mathematics · Տարրական մաթեմատիկա
LEVEL	Grade 1 · typical age 6–7
DURATION	60–72 instructional hours (approx. 60 lessons)
PREREQUISITE	Counts orally to at least 20; recognizes written numerals 0–10; can compare small groups by size.
LEADS TO	Grade 2 Mathematics (MTH-G2)

COURSE DESCRIPTION

Դասընթացի նկարագրություն

Grade 1 is the year in which a child stops counting and begins to reason about number. The course builds a secure understanding of quantity to 120, develops fluent addition and subtraction within 20 through strategy rather than memorization, and introduces place value as the organizing idea that will carry the student through all of elementary arithmetic. Students also learn to measure by iterating a unit, to tell time, to work with coins, and to describe and compose two- and three-dimensional shapes.

TEACHING APPROACH

Դասավանդման մոտեցումը Վարժարանում

At Վարժարան, Grade 1 is taught concretely before it is taught symbolically. Every new idea is met first with objects or drawings, then with a diagram, and only then with an equation — the concrete–pictorial–abstract sequence. The teacher's central job this year is not to get correct answers but to get correct explanations: a first-grader who can say why 8 + 5 is 13 is far better positioned than one who has memorized it. Lessons are 30–45 minutes, highly interactive, and paced to the child's attention rather than to the clock.

PLACEMENT AND READINESS INDICATORS

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The placement assessment checks each indicator below. A student missing two or more is enrolled with a remediation plan attached to the personalized curriculum.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Counts to 20 without skipping or repeating	Rote counting is the foundation for counting-on strategies; if absent, begin with Unit 1 extended.

READINESS INDICATOR	WHAT IT TELLS THE TEACHER
Matches a spoken number to a set of that many objects	Demonstrates one-to-one correspondence and cardinality — the true prerequisite for addition.
Recognizes and writes numerals 0–10	If reversals are present, treat as normal at this age; do not delay content.
Identifies which of two small groups has more	Comparison language ('more', 'fewer', 'the same') precedes the symbols >, <, =.

COURSE LEARNING OUTCOMES

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On completion of this course the student will be able to:

1. Count, read, write, and represent whole numbers to 120, starting from any number.
2. Add and subtract fluently within 20 using at least two reliable strategies, and within 10 from memory.
3. Explain why an addition or subtraction strategy works, using objects, drawings, or a number line.
4. Understand two-digit numbers as tens and ones, and compare them using >, <, and =.
5. Add a two-digit number and a multiple of ten, and mentally find 10 more or 10 less.
6. Solve one-step word problems of all three types (add-to, take-from, put-together/take-apart) within 20.
7. Measure length by iterating a shorter unit, and order three objects by length.
8. Tell and write time to the hour and half-hour on analog and digital clocks.
9. Identify, describe, compose, and partition basic 2-D and 3-D shapes; partition into halves and fourths.

COURSE UNITS

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Each unit below lists the lesson-level topics a teacher delivers, the objectives that define success, the observable mastery criteria used to update the student's progress profile, and the misconceptions this unit reliably produces together with the recommended teacher response.

UNIT 1 · 7-8 LESSONS

Number Sense to 20

Թվային պատկերացում մինչև 20

TOPICS COVERED

- Counting forward and backward from any number to 20
- One-to-one correspondence and cardinality
- Subitizing: recognizing 1–6 without counting
- Ten-frames and the anchor numbers 5 and 10
- Ordinal position: first through tenth
- Comparing quantities: more, fewer, equal
- Writing numerals 0–20 and matching numeral to quantity

LEARNING OBJECTIVES

1. Count forward to 20 beginning at any given number, not only at 1.
2. State how many objects are in a set without recounting (cardinality).
3. Instantly recognize arrangements of up to 6 objects without counting.
4. Decompose numbers to 10 in every way (e.g., 7 as 6+1, 5+2, 4+3) using a ten-frame.
5. Compare two sets and justify which is greater using matching, not counting.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Counts on from a given number in 4 of 5 trials without restarting at 1.
- Names 'how many' after a single count, with no recount, in 4 of 5 trials.
- Produces all pairs summing to 10 within 60 seconds using a ten-frame.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child recounts the set each time they are asked 'how many?'	Cardinality is not yet established. Cover the set after counting and ask again; use the 'hiding game' daily until the last count word is understood as the total.
The child counts every object starting from one when adding.	Introduce counting-on explicitly: put the larger number 'in your head' and count up the smaller on fingers. Do not allow the strategy to be discovered by chance.

TEACHING NOTE

This unit sets the tone for the year. Resist the urge to move quickly — a student who leaves this unit without cardinality and subitizing will compensate with finger-counting for years.

UNIT 2 · 8–9 LESSONS

Addition and Subtraction Within 10

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TOPICS COVERED

- The meaning of addition: joining and part-part-whole
- The meaning of subtraction: taking from, comparing, missing part
- The symbols +, −, = and reading an equation aloud
- Counting-on from the larger addend
- Number bonds and fact families
- Doubles and near-doubles to 10
- The commutative property discovered, not stated
- Adding and subtracting zero

LEARNING OBJECTIVES

1. Represent an addition or subtraction situation with objects, a drawing, and an equation.
2. Use counting-on rather than counting-all for sums within 10.
3. Produce the full fact family from a number bond (e.g., 3, 4, 7).
4. Know from memory all sums of two one-digit numbers within 10.
5. Explain that a + b = b + a using a picture, without using the word 'commutative'.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Recalls all facts within 10 with under 3 seconds of hesitation.
- Writes both addition and both subtraction equations for a given number bond.
- Uses counting-on, not counting-all, as the default strategy in observation.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child reads '=' as 'the answer comes next'.	This is the single most consequential misconception in elementary mathematics. Deliberately write 7 = 3 + 4 and 3 + 4 = 2 + 5. Ask 'is this true?' rather than 'what is the answer?'
Subtraction is understood only as 'take away'.	Teach all three meanings from the start — take-away, comparison, and missing addend — or the child will be unable to solve 3 + ? = 8.

TEACHING NOTE

Fluency here means efficient and flexible, not fast and memorized. A child who computes 6 + 3 by counting on from 6 is fluent; one who has memorized it but cannot explain it is not.

UNIT 3 · 9-10 LESSONS

Addition and Subtraction Within 20

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TOPICS COVERED

- Teen numbers as ten and some ones
- Making ten to add (8 + 5 = 8 + 2 + 3)
- Doubles and doubles-plus-one within 20
- Subtracting through ten (15 − 7 = 15 − 5 − 2)
- Missing-addend problems: 9 + ? = 14
- Adding three whole numbers
- The associative property in service of strategy
- Relating addition and subtraction to check work

LEARNING OBJECTIVES

1. Decompose an addend to make a ten, and explain why this does not change the sum.
2. Add and subtract within 20 using a chosen strategy, and defend the choice.
3. Solve missing-addend equations by relating them to subtraction.
4. Add three one-digit numbers by first finding a pair that makes ten.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Solves any sum within 20 in under 5 seconds using an efficient strategy.
- Explains the make-ten strategy aloud with a ten-frame or number bond.
- Correctly solves 8 of 10 missing-addend problems.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child breaks apart the wrong addend when making ten.	Always break the number that is *not* near ten. Practice deciding which addend to keep whole before computing anything.
Subtraction is performed by counting back one at a time from 15.	Slow and error-prone. Model subtracting to ten first, then the remainder — the mirror of make-ten.

TEACHING NOTE

This is the highest-leverage unit of Grade 1. Fluency within 20 is the arithmetic bottleneck for Grades 2–4. Do not advance until it is secure, even if the calendar says otherwise.

UNIT 4 · 7-8 LESSONS

Place Value to 120

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TOPICS COVERED

- Bundling ten ones into one ten
- Two-digit numbers as tens and ones
- Reading, writing, and representing numbers to 120
- Counting by tens from any number
- 10 more and 10 less, mentally
- Comparing two-digit numbers with >, <, =
- Adding a two-digit number and a multiple of ten
- Adding a two-digit number and a one-digit number

LEARNING OBJECTIVES

1. Explain that the 4 in 47 means four tens, using bundled objects.
2. Write any number to 120 given in words or in tens-and-ones form.
3. Find 10 more or 10 less than a two-digit number without counting.
4. Compare two two-digit numbers by reasoning about tens first, then ones.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Represents any number to 100 with tens and ones in two different ways.
- Answers '10 more than 63' in under 3 seconds.
- Compares two-digit numbers correctly in 9 of 10 trials, with justification.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child says 47 is 'a four and a seven'.	Place value has not been constructed. Return to physical bundling — ten straws banded together — before any symbolic work.
The child says 19 is greater than 21 because 9 > 1.	Compare tens first. Make this an explicit spoken routine: 'How many tens? Two tens beats one ten.'

TEACHING NOTE

Place value is the concept that most reliably predicts later arithmetic success. Ten minutes of bundling with real objects is worth an hour of worksheets.

UNIT 5 · 6-7 LESSONS

Measurement, Time, and Money

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TOPICS COVERED

- Comparing and ordering objects by length
- Measuring by iterating a shorter unit end to end
- Indirect comparison: if A > B and B > C, then A > C
- Telling time to the hour and half-hour
- Analog and digital clock faces
- Identifying coins and their values (penny, nickel, dime, quarter)
- Counting collections of like coins

LEARNING OBJECTIVES

1. Order three objects by length and justify using direct comparison.
2. Measure an object with same-size units placed with no gaps or overlaps.
3. Tell and write time to the half-hour on both clock types.
4. Identify each US coin and state its value.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Measures accurately with no gaps or overlaps in 4 of 5 trials.
- Reads half-past times correctly in 8 of 10 trials.
- Names all four common coins and their values without prompting.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
The child leaves gaps between units or overlaps them when measuring.	Emphasize that measurement units must tile the object exactly. Let the child measure the same object twice, carelessly and carefully, and compare results.
The child reads 3:30 as 'four-thirty' because the hour hand is near the 4.	Teach the hour hand as moving continuously between numbers; it names the hour it has *passed*, not the one it approaches.

TEACHING NOTE

Money is included because it makes place value tangible: a dime is a bundled ten. Use coins deliberately when reviewing Unit 4.

UNIT 6 · 6-7 LESSONS

Geometry and Fractions as Fair Shares

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TOPICS COVERED

- Defining vs. non-defining attributes of shapes
- Naming 2-D shapes: triangle, rectangle, square, trapezoid, circle, hexagon
- Naming 3-D shapes: cube, cone, cylinder, sphere, rectangular prism
- Composing new shapes from smaller shapes
- Partitioning circles and rectangles into two and four equal shares
- The language of halves, fourths, and quarters
- Understanding that more equal shares means smaller pieces

LEARNING OBJECTIVES

1. Distinguish attributes that define a shape (number of sides) from those that do not (color, size, orientation).
2. Build a larger shape from pattern-block pieces and describe the composition.
3. Partition a shape into two or four equal shares and name one share.
4. Explain why a fourth of a pizza is smaller than a half of the same pizza.

MASTERY CRITERIA — EVIDENCE REQUIRED TO MARK THIS UNIT COMPLETE

- Identifies a triangle regardless of orientation or proportion in 5 of 5 trials.
- Partitions a rectangle into four equal shares and names one share correctly.
- States that fourths are smaller than halves, with a drawing to justify.

COMMON MISCONCEPTIONS AND TEACHER RESPONSE

MISCONCEPTION	RECOMMENDED RESPONSE
A triangle resting on a vertex is not recognized as a triangle.	Prototype thinking. Show non-standard orientations and irregular triangles from the very first lesson.
Any two pieces are called 'halves' regardless of size.	Halves must be *equal* shares. Deliberately cut a paper unequally and ask whether the child wants the 'half' on the left.

TEACHING NOTE

Fair-shares language planted here is what makes Grade 3 fractions feel familiar rather than foreign. Say 'one of two equal shares' before ever writing 1/2.

ASSESSMENT AND MASTERY POLICY

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ASSESSMENT	WHEN	WHAT IT MEASURES
Placement Assessment	Before enrollment	Counting, cardinality, numeral recognition, comparison; produces the initial academic profile.
Unit Check	End of each unit · 10–15 min	Mastery criteria for that unit only; oral and written, teacher-administered live.
Fluency Probe	Every 3rd lesson · 3 min	Addition/subtraction facts within 10, then within 20; tracks strategy use, not just accuracy.
Mid-Year Assessment	After Unit 3	Cumulative Units 1–3; triggers pacing adjustment if below 80%.
End-of-Course Assessment	After Unit 6	All course outcomes; determines readiness for Grade 2 placement.

Վարժարան reports mastery, not grades. Each topic in the student's profile is marked **Mastered**, **Developing**, or **Needs Review** against the criteria stated in each unit above. A topic is marked Mastered only when the criterion is met on two separate occasions at least one week apart — this prevents short-term recall from being recorded as durable learning. Topics marked Needs Review are automatically re-entered into homework and into the opening minutes of subsequent lessons until they are re-mastered.

PACING OPTIONS

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TRACK	CADENCE	EXPECTED COMPLETION
Standard	1 lesson/week, 45 min	Full course in approximately 40–45 weeks (one academic year).
Accelerated	2 lessons/week, 45 min	Full course in approximately 22–25 weeks.
Intensive / Catch-Up	2–3 lessons/week	Used when the placement assessment shows the student is behind grade level; 16–20 weeks.
Summer Program	3 lessons/week, 6 weeks	Units 1–3 only, or targeted remediation of identified gaps.

Pacing is assigned after the placement assessment and reviewed at the mid-year assessment. A student who is meeting mastery criteria early may be moved to the accelerated track; a student who is not may be moved to intensive without any change in the course content itself.

HOMEWORK AND INDEPENDENT PRACTICE

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Grade 1 homework is short by design: 10 minutes, no more than five problems, assigned after every lesson. Its purpose is retrieval practice, not new learning — nothing is assigned that was not taught. Roughly one third of each assignment revisits earlier units so that fluency is maintained rather than lost. Parents are asked to sit nearby but not to teach; if the child is stuck for more than two minutes, the correct action is to stop and note it for the teacher.

PROGRESS MILESTONES REPORTED TO PARENTS

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These are the achievements the parent portal announces as the student reaches them. They replace attendance counts as the primary measure of progress.

- Understands that the last number counted tells how many (cardinality).
- Knows all addition and subtraction facts within 10 from memory.
- Adds and subtracts fluently within 20 using the make-ten strategy.
- Reads, writes, and compares any number to 120.
- Finds 10 more or 10 less than any two-digit number mentally.
- Tells time to the half-hour and identifies all US coins.
- Partitions shapes into halves and fourths and explains why fourths are smaller.
- Completed Grade 1 Mathematics — ready for Grade 2.

MATERIALS AND RESOURCES

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- Digital whiteboard with movable counters, ten-frames, and a number line to 120 (teacher-shared).
- Physical set at home: 100 counting objects, ten-frame cards, a set of US coins, a small ruler.
- Pattern blocks (physical or on-screen) for the geometry unit.
- Վարժարան Grade 1 problem bank — approximately 600 items tagged by unit and difficulty.

Վարժարան · Varzharan — Exceptional Armenian educators. American academic standards. Personalized education. | This syllabus defines the standard course. Every enrolled student receives a personalized version of it, sequenced from their placement assessment and revised as their progress profile changes.